

Artificial Intelligence(H)

Mock Final Exam

Southern University of Science and Technology

Spring 2026

Time: 120 minutes

Total: 100 points

Instructions: Show necessary derivations. For search and pruning questions, follow the specified order.

This mock paper follows the style of the reconstructed final exam. All graphs, trees, and datasets are provided in text form so that the paper is self-contained.

Problem 1: Search

[15 pts]

Consider the following undirected graph. Every edge has uniform cost $c > 0$.

Node	Neighbors in alphabetical order
<i>A</i>	<i>B, C, D</i>
<i>B</i>	<i>A, E, F</i>
<i>C</i>	<i>A, D, F, G</i>
<i>D</i>	<i>A, C, H</i>
<i>E</i>	<i>B, I</i>
<i>F</i>	<i>B, C, G, I</i>
<i>G</i>	<i>C, F, H, I</i>
<i>H</i>	<i>D, G</i>
<i>I</i>	<i>E, F, G</i>

The start node is *A* and the goal node is *G*. If there are multiple nodes to expand, always use alphabetical order.

(a) Draw the breadth-first search tree until the goal is found. State the expansion order. [5 pts]

(b) Now let $c = 1$ and use the following heuristic function:

n	A	B	C	D	E	F	G	H	I
$h(n)$	4	3	2	3	3	1	0	2	1

Draw the A* search tree until G is selected for expansion. For each generated node, write $g(n)$, $h(n)$, and $f(n) = g(n) + h(n)$. **[7 pts]**

(c) For the same heuristic table, find the range of c that guarantees admissibility. Show your calculation. **[3 pts]**

Problem 2: Minimax and Alpha-Beta Pruning

[13 pts]

Consider the following game tree. The root is a MAX node. Children are visited from left to right.

Root child	Second-level child	Leaf utilities
A (MIN)	A_1 (MAX)	3, 12
	A_2 (MAX)	8, 2
	A_3 (MAX)	4, 6
B (MIN)	B_1 (MAX)	14, 5
	B_2 (MAX)	2, 1
	B_3 (MAX)	7, 3
C (MIN)	C_1 (MAX)	6, 9
	C_2 (MAX)	4, 0
	C_3 (MAX)	10, 11

(a) Compute the minimax value of every internal node. **[4 pts]**

(b) Run alpha-beta pruning and write the final α and β values for each visited internal node. If a node is pruned, write NA. **[5 pts]**

(c) Mark which branches are pruned. **[2 pts]**

(d) Mark the final path chosen by MAX. **[2 pts]**

Problem 3: CSP and AC-3

[12 pts]

(a) State the worst-case time complexity of AC-3 and briefly explain why. **[4 pts]**

(b) Consider a CSP with variables X, Y, Z :

$$D_X = D_Y = D_Z = \{1, 2, 3, 4\}, \quad C = \{X < Y, Y < Z\}.$$

Use AC-3 to enforce arc consistency. Show each domain pruning step and the final domains. **[8 pts]**

Problem 4: Logic and CNF**[12 pts]**

- (a) Convert the following propositional formula into CNF:

$$(p \wedge q) \rightarrow \neg(p \leftrightarrow r).$$

Simplify the result as much as possible.

[5 pts]

- (b) Four robots R_1, R_2, R_3, R_4 may be selected for a mission. Let x_i mean robot R_i is selected. Represent the following rules in propositional logic and convert them into CNF.

[5 pts]

(i) R_1 and R_3 must be selected together.

(ii) Exactly one of R_2 and R_4 is selected.

(iii) If R_2 is selected, then R_1 must also be selected.

- (c) Suppose additionally x_2 is true. Use unit propagation to derive which of x_1, x_3, x_4 must be true or false.

[2 pts]**Problem 5: Perceptron and Logistic Regression****[10 pts]**

- (a) Write the perceptron prediction rule and weight update rule for labels $y \in \{-1, +1\}$.
- (b) Given $w = (-1, 1)^T$, $b = 0$, learning rate $\eta = 0.5$, and one training example $x = (2, 1)^T$, $y = +1$, perform one perceptron update if needed.
- (c) Derive the gradient descent update for binary logistic regression with cross-entropy loss:

[3 pts]**[3 pts]**

$$J(w, b) = -\frac{1}{m} \sum_{i=1}^m [y_i \log h_i + (1 - y_i) \log(1 - h_i)], \quad h_i = \sigma(w^T x_i + b).$$

[4 pts]**Problem 6: SVM****[10 pts]**

Consider two one-dimensional data points $(x_1, y_1) = (2, +1)$ and $(x_2, y_2) = (-2, -1)$.

- (a) Write the primal objective function and constraints of the soft-margin SVM. Use C as the penalty parameter.
- (b) For these two points, write the constraints explicitly in terms of w, b, ξ_1, ξ_2 .
- (c) Suppose the optimal w is given by $w^* = \min(C, \frac{1}{2})$. Find the possible optimal value(s) of b for the cases $C \geq \frac{1}{2}$ and $0 < C < \frac{1}{2}$.

[4 pts]**[2 pts]****[4 pts]****Problem 7: Naive Bayes****[12 pts]**

The following dataset has four binary features and one binary label.

ID	X_1	X_2	X_3	X_4	Y
1	1	0	1	0	+
2	1	1	1	0	+
3	0	1	0	1	−
4	0	0	0	1	−
5	1	0	0	0	+
6	0	1	1	1	−
7	1	1	0	0	+
8	0	0	1	1	−

Use Bernoulli Naive Bayes with Laplace smoothing $\alpha = 1$.

- (a) Compute the class priors $P(Y = +)$ and $P(Y = -)$. [2 pts]
- (b) Compute all conditional probabilities needed for prediction, i.e. $P(X_j = 1 \mid Y)$ and $P(X_j = 0 \mid Y)$ for all j . [5 pts]
- (c) Predict the label of $x = (1, 0, 1, 0)$. Show both class scores. [3 pts]
- (d) If the Naive Bayes independence assumption is not used, how many feature-label combinations must be considered for four binary features and one binary label? How many independent parameters does the full joint distribution over the five binary variables have? [2 pts]

Problem 8: Bayesian Networks and Sampling

[14 pts]

Suppose $A \rightarrow B \rightarrow C \rightarrow D$ is a Bayesian network with:

$$P(A) = \frac{1}{4}, \quad P(B \mid A) = \frac{1}{5}, \quad P(B \mid \neg A) = \frac{3}{4},$$

$$P(C \mid B) = \frac{1}{2}, \quad P(C \mid \neg B) = \frac{1}{3}, \quad P(D \mid C) = \frac{1}{6}, \quad P(D \mid \neg C) = \frac{7}{8}.$$

- (a) Factorize the joint distribution $P(A, B, C, D)$ according to the network. [2 pts]
- (b) Given these prior samples:

$(A, B, \neg C, \neg D), \quad (A, \neg B, C, \neg D),$
 $(\neg A, B, C, \neg D), \quad (\neg A, \neg B, C, \neg D),$
 $(A, \neg B, \neg C, D), \quad (A, B, C, \neg D),$
 $(\neg A, B, \neg C, D), \quad (\neg A, \neg B, C, \neg D),$

 estimate $P(C)$ by prior sampling. [2 pts]
- (c) Use rejection sampling on the same samples to estimate $P(C \mid A, \neg D)$. State which samples are kept. [3 pts]

- (d) For likelihood weighting to estimate $P(\neg A \mid B, \neg D)$, compute the weight of each sample:

$$\neg A, B, C, \neg D; \quad A, B, C, \neg D; \quad A, B, \neg C, \neg D; \quad \neg A, B, \neg C, \neg D.$$

[4 pts]

- (e) From the four weighted samples in part (d), estimate $P(\neg A \mid B, \neg D)$.

[3 pts]

Problem 9: Genetic Algorithm and Course Projects

[12 pts]

- (a) Suppose we evolve the weights of a neural network using a genetic algorithm instead of backpropagation. Describe how to set up a fair experiment comparing these two training methods. [5 pts]
- (b) In Project 1, the objective of information exposure maximization can be estimated by Monte Carlo simulation. Explain what the state/individual, fitness function, and stopping criterion could be for a GA-based solver. [4 pts]
- (c) In Project 3, a recommender system is evaluated by AUC and nDCG@5. Briefly explain what each metric measures and why a model may improve one but not the other. [3 pts]